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Beyond blood pressure: pulse wave analysis – a better way of assessing cardiovascular risk?

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The study of the pulse pressure wave using the technique of applanation tonometry is becoming more popular as a method of studying vascular stiffness. This review aims to present the uses, potential uses, strengths and weaknesses of the technique of applanation tonometry to assess augmentation index and pulse wave velocity. This article shall briefly discuss the practicalities of applanation tonometry, the physiological factors affecting pulse wave velocity and pulse wave analysis, the changes in pulse wave velocity and pulse wave analysis with pharmacological interventions, and the use of applanation tonometry as a prognostic tool. Although the technique of applanation tonometry seems initially promising, several pertinent issues need to be addressed before it can be used reliably as a clinical tool. Importantly, the technique of applanation tonometry to derive the central waveform from noninvasively-acquired peripheral data needs to be prospectively validated. As stated in this review, the transfer function has been extensively validated using invasive data, whilst noninvasive data estimation of the ascending aortic blood pressure is less accurate.

Keywords: applanation tonometry, augmentation index, pulse pressure, pulse wave velocity

future medicine

Blood pressure (BP) measurement in the brachial artery has long been performed to give an indication of cardiovascular risk. However, brachial artery BP measurement has its limitations; the brachial artery does not suffer from atherosclerosis and the BP recorded in this artery is removed from the BP in the ascending aorta, which is the BP that most directly loads the heart. Advances in technology now allegedly allow the ascending aortic BP (amongst other vascular parameters) to be recorded using the technique of pulse pressure wave analysis (PWA). This has led many scientists and clinicians to postulate that assessment of the pulse pressure wave should be a better way of assessing risk than simply measuring brachial artery BP [1,2]. Although this is an attractive idea, we must remember that there is a strong body of evidence linking brachial artery BP to morbidity and mortality [3]. Such convincing evidence of prognostic value is currently lacking for the technique of PWA.

The simplicity of the technique of assessing the peripheral pulse waves and portability of the equipment are obvious incentives for its use as a research tool. It may also become established as a powerful clinical tool if it proves its worth in longitudinal studies. The method of PWA to derive the central aortic pressure waveform, and the method of calculating pulse wave velocity (PWV) using applanation tonometry shall be briefly discussed. The authors shall then assess the evidence currently available to help us to decide whether these methods can add anything to BP measurement as a clinical tool for assessing and treating disease.

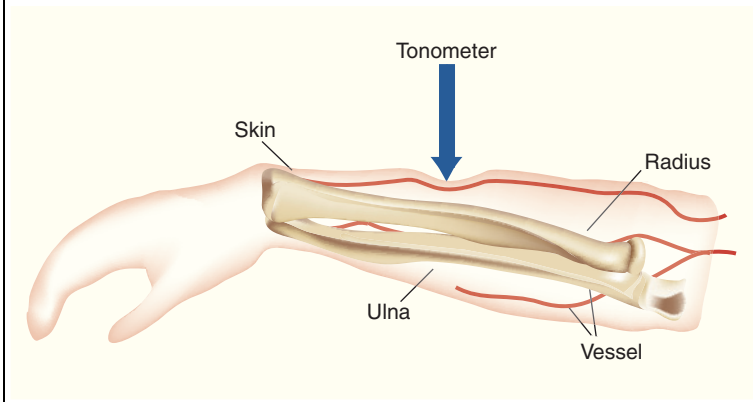
There are two main methods of using applanation tonometry to assess the pulse prssure waveform. It can be analyzed by applying a valid transfer function based on Fourier analysis, which allows derivation of an aortic pressure waveform, the augmentation index (AIx) and time to reflected wave [4–6]. The pulse pressure wave can also be analyzed using a modified Windkessel model of circulation, which allows calculation of large and small vessel compliance [7]. The scope of the review shall be limited to the discussion of PWA and PWV by the technique of applanation tonometry using the method based on Fourier analysis.

Methodology of pulse wave analysis

The pulse, rather than being a unit of pressure, the limits of which are defined by the upper and lower limits of BP, consists of a series of harmonics travelling along the arterial tree. The pulse pressure wave is formed from the combination of the incident wave (i.e., the pressure wave generated by the left ventricle in systole) and waves reflected back from the periphery [2].

Applanation tonometry detects this pulse pressure wave by compressing the artery between a micromanometer-tipped probe and the underlying structures (**Figure 1**). This pressure waveform is then digitalized such that it can be viewed on a computer screen.

Figure 1. Performing applanation tonometry.

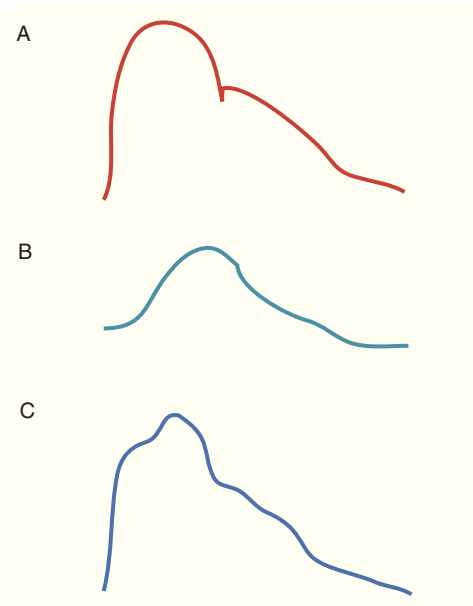


Pulse pressure waveform

The pulse pressure waveform varies in different vessels in the same subject, and is dependent on the pattern of left ventricular ejection, viscoelastic properties of the artery, the viscosity of the blood, wave reflection and wave dispersion. This waveform is altered if the blood vessel anatomy is changed, for example in aortic stenosis.

Nowadays, rather than just being able to measure pulse pressure in palpable vessels, computer software purports to allow the calculation of the central aortic pressure and wave form from the radial or carotid pulse pressure waveform, calibrated using peripheral BP. This requires the use of a mathematical equation, the 'generalized transfer function' (TF) that can be built into the computer software. Many transfer functions exist [4,8,9] and have been derived as a result of studies recording both the peripheral waveform at the same time as the central ascending aortic waveform obtained invasively.

Figure 2. Incident and reflected waves combine to produce the observed pressure.



A: Incident wave (ascending aorta); B: Reflected wave (ascending aorta); C: Resultant wave (ascending aorta).

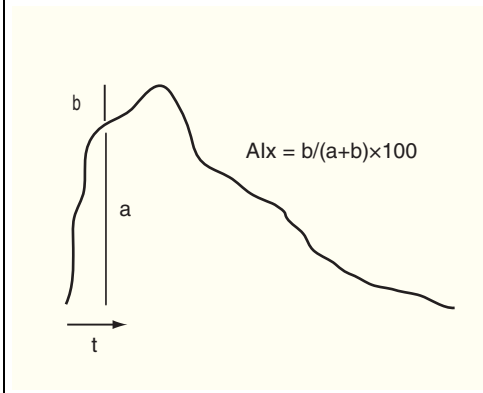
Augmentation of the pulse pressure waveform

Wave reflection from the periphery to the central blood vessels augments the aortic pressure wave. An incident wave travelling from the left ventricle (LV) to the periphery reaches smaller vessels which act as a mirror, reflecting it back to the aorta. Thus, the resulting pressure in the ascending aorta is the sum of the incident and reflected wave (**Figure 2**). The effect of the reflected wave on the incident wave in the ascending aorta is expressed as the AIx, and is a measure of the additional load that the left ventricle is subject to due to wave reflection. AIx is calculated as the height of the reflected wave as a percentage of the total height of the ascending aortic pressure wave (**Figure 3**). AIx depends on:

- The length of the cardiac cycle (which depends on the heart rate)
- The duration of systole and diastole
- The speed of the pulse wave
- The incident waveform
- The amplitude of the reflected pulse wave

For a given heart rate, the time of arrival of the reflected wave depends on the speed of the pulse (PWV), which is determined by the stiffness of the vasculature (PWV increases with increasing vessel stiffness). If the reflected wave arrives early in the cardiac cycle it combines with the incident wave, giving a greater ascending aortic pressure against which the left ventricle has to pump. However, if it arrives later in the cardiac cycle it increases the ascending aortic pressure in diastole leading to improved coronary circulation. If the cardiac cycle is longer (i.e., the heart rate is slower) the reflected wave is more likely to arrive in systole. As AIx is dependent on PWV, which in turn is dependent on arterial stiffness, it is no surprise that AIx has been proposed as a marker for arterial stiffness [10,11]. Although there is a correlation between PWV and AIx [12] the relationship between the two is not always clear.

Figure 3. Calculation of augmentation index.

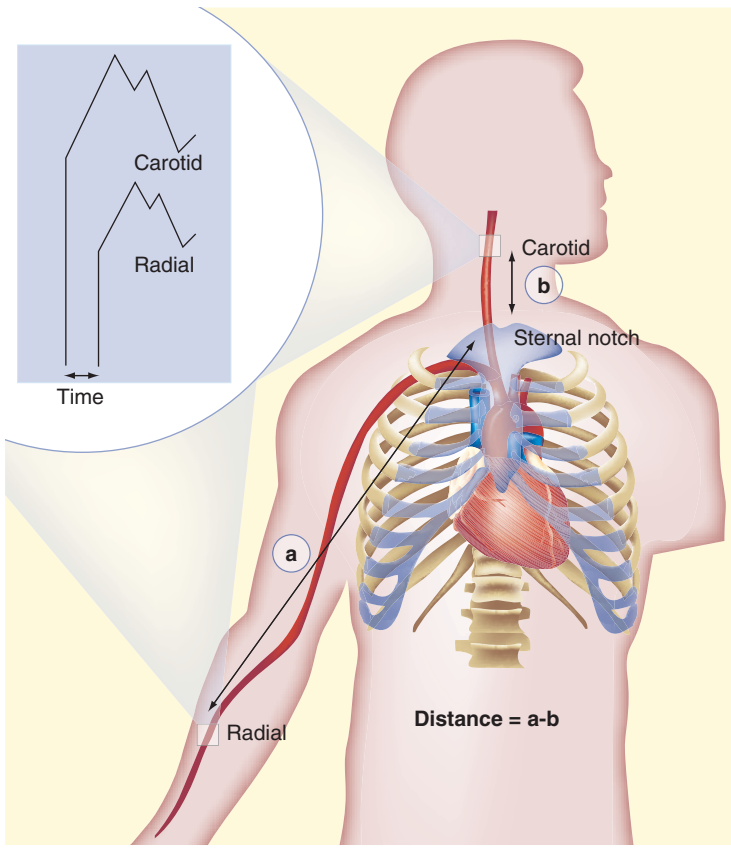


AIx: Augmentation index; T: time to reflected wave.

Pulse wave velocity

PWV and its calculation were first described by Bramwell and Hill in 1922 [13]. Velocity is calculated by dividing distance travelled by time taken to travel that distance and the results are presented in ms^{-1}.

Figure 4. How to calculate PWV.



To calculate the PWV the time delay between the pulse pressure wave at two different sites a measured distance apart has to be calculated (**Figure 4**). This can be performed either by placing a probe on each site (usually the carotid and radial, or carotid and femoral) and recording the wave forms simultaneously, or by recording the waveforms at the different sites independently but comparing the time delay at both sites to a simultaneously measured QRS complex. The method of calculation of PWV depends on the system used.

Reproducibility

In general the technique of applanation tonometry is simple to learn, and once it has been learnt there is good inter- and intra-operator reproducibility [14,15]. PWV measured using applanation tonometry has a repeatability coefficient of over 90% [16]. There are published guidelines to help to standardize measurement conditions and hence reduce longitudinal variation in PWV measurement [17].

It is also important to note that different studies have used different sites between which to assess PWV; the difference in elasticity and propensity to atherosclerosis of the arterial systems studied may cause incompatible results when comparing studies.

Validity

Assessment of AIx & ascending aortic BP

The TFs used to calculate ascending aortic BP and AIx from the radial pulse in commercially available software packages give similar results to each other, supporting their validity [4,8]. Although there have been many studies performed that have aimed to validate the transfer functions [8,9,18,19], in all of these studies some or all of the data used to calibrate the waveforms were obtained invasively.

Three studies have attempted to validate a TF to assess ascending aortic BP using non invasive measurements for calibration. In all cases the authors found the technique to be unreliable in predicting ascending aortic BP [20–22]. However, it has been argued that these difficulties are not due to the TF *per se* [23], rather to the method used to obtain BP measurements to calibrate the system. Despite this, it makes no sense to use this technique in the clinic setting if the radial artery waveforms have to be calibrated using invasively obtained BP measurements. To the authors' knowledge there are no human studies validating the AIx obtained by wholly noninvasive measurements.

Box 1. Physiological factors affecting PWA & PWV.

AIx increases with:
- Slower heart rate
- Shorter stature
- Age
- Gender
- Ejection fraction

PWV increases with: Age and BP
PWV may change with heart rate

Assessment of PWV

The authors know of no human studies comparing invasively measured PWV with that obtained noninvasively using applanation tonometry. The assumptions used in performing PWA do not apply to the assessment of PWV.

Physiological factors affecting PWA & PWV

The physiological factors impacting upon PWA and PWV are listed in **Box 1** and shall be discussed here.

Heart rate

AIx is inversely related to heart rate [24,25] as a fast heart rate shortens the ejection duration, meaning that the reflected wave arrives later in the cardiac cycle, relative to the incident wave. Therefore, with faster heart rates the reflected wave is more likely to arrive in diastole, thus the left ventricle has to work against a smaller pressure, and there is greater perfusion to the coronary arteries (2). This is contrary to our general belief that a slower heart rate is better prognostically and the knowledge that β-blockers, which slow the heart rate have great prognostic benefit in the treatment of heart failure and to a lesser extent in hypertension. Although it has been proposed that AIx should be corrected for heart rate, there is a school of thought that dictates that the change in AIx with heart rate may have pathophysiological effects that should not be adjusted for.

The relationship between PWV and heart rate is a little less clear. Some cross-sectional studies have shown no association between PWV and heart rate [12,26], whilst others have indicated that PWV does vary with heart rate [27]. Studies using pacing to alter heart rate have also yielded conflicting results [24,27]. Therefore, at the present time, caution must be exercised when interpreting the results from both PWA and PWV when using interventions that alter heart rate.

Height

AIx is inversely related to height [28]. This is assumed to be due to the shorter distance from the origin of the waveform to the point of reflection, leading to a quicker return of the reflected wave for a given PWV. It has been suggested that height should be controlled for in the analysis of results [28]. Conversely, it may be that the increase in augmentation seen in shorter people contributes to the genuine increase in cardiovascular risk seen with decreasing height [28,29].

Aging

Arteries stiffen as people age, which is reflected in an increase in AIx and PWV that may be increased out of proportion with the age related increase in BP [30,31], suggesting that aging-related changes in the vasculature can be underestimated by assessment of BP [30].

Gender

AIx is higher in women than in men and this difference is not explained by the difference in height between the sexes [32,33].

Cross-sectional studies

If risk factors for cardiovascular disease and atherosclerotic load are associated with increased PWV and AIx this would strengthen the case for using PWV/AIx as markers of atherosclerotic risk.

Arterial stiffness assessed by a variety of methods has been correlated with insulin, cholesterol and triglyceride levels, hypertension and smoking [34–38]. PWV correlates with the number of (treated and untreated) cardiovascular risk factors and atherosclerotic events, and is increased in both Type 1 and Type 2 diabetes [39,40]. The evidence is more uncertain when considering the change in AIx in diabetes [41,42]; however, AIx is known to be associated with the presence and severity of coronary artery disease [43]. Both PWV and AIx correlate with cardiovascular risk assessed by the Framingham risk equations and ESC risk chart, respectively [44–46].

PWV is also positively correlated with carotid intima-media thickness (IMT), a marker, not only of atherosclerotic burden in the carotid artery, but also a reflection of that in the coronary arteries [47,48]. Perhaps more importantly PWV correlates with coronary artery plaque burden [49]. It is perhaps most interesting that PWV and AIx are associated in both univariate and multivariate analyses with time to ischemia on ETT in patients with

Table 1. Intervention and effect of intervention on PWV and AIx.

Intervention	PWV	AIx
ACE inhibition	↓	↓
β-blockade	?	?
AII blockade	↓	↓
Renal transplant in ESRD [55]	↓	↓
Statins	↓	?
Antioxidants	↓	↓

Intervention either increases (↑) or decreases (↓) parameters, or the evidence is lacking or equivocal (?). AIx: Augmentation index; ESRD: End-stage renal disease; PWV: Pulse wave velocity.

known coronary artery disease (CAD) [50], thus potentially identifying more advanced CAD or even stable as opposed to unstable CAD.

Increased left ventricular mass (LVM) is well known to be a poor prognostic marker, thus indices that are strongly and positively associated with increasing left ventricular mass index (LVMI) are also likely to confer a similarly poor prognosis. It would be logical that LVM would be affected to a great extent by the pressure in the ascending aorta, indeed a greater estimated AIx is associated with a greater LVMI [48]. Although LVMI correlates with PWV in univariate analysis, in multivariate analysis the effect of PWV on LVMI can be accounted for by the effect of BP [51]; this is not really surprising as PWV is known to be influenced by BP [52]. It may be that AIx provides a better reflection of systemic vascular stiffness than PWV since AIx is affected by the amplitude of the reflected wave as well as its velocity.

Changes in PWA & PWV parameters with interventions

If AIx and PWV are found to improve with beneficial interventions and worsen with interventions that are harmful, this would help to support the case being made for them being useful clinical measures either alone or in combination with other cardiovascular risk factors.

Most treatments that reduce BP lead to a reduction in AIx and PWV (**Table 1**) [53,54], however this article shall focus on β-blockers or renin–angiotensin–aldosterone system blockers as they have been recently investigated as agents that may also modify mortality independent of their antihypertensive effects.

Angiotensin-converting enzyme nhibition

Angiotensin-converting enzyme (ACE) inhibitors have well-known beneficial cardiovascular effects [56] and have been shown to reduce AIx and PWV, assessed by applanation tonometry in hypertensive patients [57,58].

It has been suggested that the reduction in LVM seen with ACE inhibition may be related to the decrease in PWV, but it is not known if the reduction in LVM is a direct result of the reduction in PWV. If this were the case this would lend further weight to the argument that indices of vascular stiffness are useful for gauging cardiovascular risk.

β-blockade

β-blockers improve morbidity and mortality outcomes in patients with heart failure [59–61] and reduce LVM in those with hypertension [62]. Stimulating the sympathetic nervous system with noradrenaline (NA), results in an increase mean arterial pressure (MAP), PWV and AIx, which would be consistent with it exerting adverse effects on the vascular system [63]. Furthermore, some β-blockers (e.g., bisoprolol) do improve vascular stiffness as demonstrated by a decrease in PWV [64]. Unfortunately, this short-term effect on PWV has not been shown to be maintained in the long term [65].

However, β-blockers decrease heart rate, which, as prevoiusly stated, leads to an increase in AIx, which is thought to be deleterious. It may be that different types of β-blocker have varying effects on vascular stiffness due to differing receptor affinities. For example, some β-blockers lead to an increase in AIx by decreasing heart rate, whilst other β-blockers that have vasodilating properties leading to a decrease in AIx [66]. β-blockers also prolong diastole, thus the increase in AIx seen with some β-blockers may be offset by a beneficial effect on coronary artery perfusion.

The data thus far with respect to β-blockers may suggest either that AIx and PWV are relatively unimportant measures of outcome response, which is possible since the benefit seen with β-blockers may relate to their myocardial protective actions rather than to their vascular effects. However, it may be that selecting β-blockers that have a beneficial effect on PWV and AIx will lead to an even better long-term outcome.

Other antihypertensive drugs

Some antihypertensive drugs have greater effects on the arterial tree than others independent of the magnitude of BP reduction [65]. For example, although fosinopril and atenolol reduce

ambulatory BP to a similar degree, fosinopril has a much greater beneficial effect on the AIx than atenolol [67]. Although the apparent disadvantage of atenolol could be due to the decrease in heart rate seen with this drug.

It is of great interest here that the Losartan Intervention For Endpoint (LIFE) study has shown losartan to be of greater benefit than atenolol with respect to the combined end point of cardiovascular events for the same reduction in BP [68,69]. When this end point is broken down it is seen that the cause of this improved outcome with losartan is due to a reduction in cerebrovascular events, rather than cardiovascular events. It may be that the main protective effect of atenolol is due to its myocardial protective, anti-ischemic effects, whereas losartan has greater vascular protective effects on both the coronary arteries and on the cerebrovasculartre. Although losartan decreases PWV and AIx [70,71], its effects on these parameters has not been compared with atenolol. However, an earlier study comparing the effects of atenolol and fosinopril on AIx certainly makes this theory attractive [67].

Statins

The benefits of statins in primary and secondary prevention of cardiovascular disease [72] may be partially attributed to the improvement in blood vessel stiffness, reflected in a decrease in PWV that they produce [73]. The decrease in aortic stiffness seen with statins may also contribute to the improvement in cardiac function observed with statins in patients with impaired left ventricular function [74]. There are no studies as yet demonstrating that statin therapy improves AIx.

Antioxidants

Antioxidants e.g., vitamins E and C and allopurinol (a xanthine oxidase inhibitor) have potential beneficial effects on the cardiovascular system. The main reason behind this is thought to be due to an improvement in endothelial function [75] secondary to decreased nitrogen oxide (NO) degradation by free radicals. Unfortunately, the evidence of benefit in large outcome trials is mixed [56,76,77]. Therefore, the fact that vitamin E has been shown to improve arterial compliance, and vitamin C is associated with a decrease in AIx may be arguments against using PWA/PWV as indicators of the value of novel therapies [78–80].

Longitudinal Studies

The strongest evidence for the use of PWA/PWV for clinical or research use will only come from longitudinal studies. Only if these studies show that AIx or PWV are predictive of future events and are either better than, or a useful addition to traditional cardiovascular risk factors will it be justifiable to use them in clinical practice.

The body of evidence for using PWV to predict mortality is becoming increasingly convincing, unfortunately the evidence for AIx in longitudinal studies is lacking (Table 2). PWV was the strongest predictor of cardiovascular mortality in a cohort of elderly patients [81]. In patients with end-stage renal disease (ESRD) both AIx and PWV are independent predictors of mortality and have a greater predictive power than pulse pressure alone [82,83]. Perhaps more importantly, PWV is significantly associated with the occurrence of coronary events in hypertensive patients even after adjusting for the Framingham score or classic risk factors [84]. Futhermore, PWV is a better predictor of mortality than BP in a cohort of Type 2 diabetic and impaired glucose tolerance patients [39]. This suggests that PWV measurement may become a useful tool in assessing risk in hypertensive patients either on its own or in addition to commonly used risk factors and risk factor profiles.

Conclusion

Noninvasive assessment of the pulse wave using applanation tonometry to measure PWV and AIx is quick and easy to perform and requires a minimum amount of training. Whilst it seems that PWV may become a useful tool for assessing cardiac risk the evidence for the use of AIx in this field is currently lacking.

Future perspective

Although assessment of the pulse pressure wave has been carried out for many years, modern day assessment using computer technology is still, relatively, in its infancy. The next 5 to 10 years should demonstrate a proliferation of research into the uses of this technology and will

Table 2. Shows patient group and whether PWV or AIx have prognostic value or the evidence is lacking.

Subject group	PWV	AIx
Hypertensive	✓	?
Elderly	✓	?
ESRD	✓	✓
Type 2 diabetes	✓	?

AIx: Augmentation index; ESRD: End-stage renal disease; PWV: Pulse wave velocity; ✓: Prognostic value; ?: lacking evidence.

hopefully allow us to decide whether it has a place in clinical practice.

Before these measurements enter general clinical practice to aid in risk stratification, they need to be properly assessed in large longitudinal follow-up studies involving different groups of patients. They may have a second possible use in the screening of promising new therapies; if treatment-induced changes in PWV and AIx parallel changes in cardiovascular events, then PWV and AIx could be very useful in screening new therapies. Although this technique seems promising it will have to be shown to be better than currently used cardiac risk factor profiles or to add significantly to them in order for it to enter clinical practice.

Executive summary

Introduction

- The analysis of the pulse pressure wave is undergoing a renaissance as the development of computer equipment means that pulse pressure wave analysis is simple to perform. Pulse pressure wave analysis allows calculation of the speed of the pulse (PWV) and "augmentation" of the pulse pressure in the ascending aorta (AIx), both of these measures are related to vascular stiffness, a key component of atherosclerosis.
- AIx reflects the extra pressure experienced by the left ventricle due to reflection of the pulse pressure wave from the periphery.
- Central aortic blood pressure (BP) can also be estimated.

Methodology of pulse pressure wave analysis

- The pulse pressure wave can be assessed by applanation tonometry, compressing the artery against structures beneath it using a micromannometer tipped probe.
- The equipment is simple to use and portable.
- The parameters mentioned above are calculated by computer software.

Validity and reproducibility

- The measurements obtained are reproducible.
- Estimation of ascending aortic BP using entirely non invasively obtained data may not be accurate.
- There are no human studies known that have compared invasive measurements of PWV and AIx to those obtained by applanation tonometry and the appropriate computer software.

Physiological factors affecting AIx and PWV

- AIx depends on length of cardiac cycle; treatments that slow heart rate (e.g., β-blockers) increase AIx.
- AIx also increases with shorter stature, age, gender, ejection fraction.
- PWV increases with age and BP and may change with heart rate.

Pathological factors affecting AIx and PWV

- AIx and PWV increase with cardiac risk factors, known coronary artery disease and the Framingham risk score.

Do AIx and PWV change with interventions that improve prognosis?

- In most studies PWV and AIx improve with interventions that improve outcome (the exception is ß-blockers, which improve outcome, but increase AIx).

Do AIx and PWV indicate prognosis in a prospective fashion?

- An increase in PWV is associated with a worse prognosis in hypertensive patients, the elderly, patients with end stage renal disease (ESRD) and those with Type 2 diabetes.
- The evidence for use of PWV is more convincing than for AIx at present time, however, AIx indicates prognosis in patients with ESRD.

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